Julia, a programming language for Operations Research

Prof. Marc Sevaux

Lab-STICC – Université de Bretagne-Sud – Lorient – FRANCE

with the support of Dr. Alexandru OLTEANU, Quentin PERRACHON and Owein THUILLIER

April 27-29, 2026

echo $(whoami)

http://people.univ-ubs.fr/marc.sevaux/

- studied in the **Applied Mathematics Institute** (Angers, France)
- got a **Master and PhD degree** from Paris VI university, France
- **associate prof.** 6 years at univ. of Valenciennes, France
- since 2005, **professor** at univ. Bretagne Sud (Lorient, France)
- **sabbatical** 2010/2011 in Helmut Schmidt univ. (Hamburg, Germany)
- EURO President 2020-2023 `http://www.euro-online.org`

- research topics: heuristics, metaheuristics, MP, matheuristics
- interest: combining metaheuristics and machine learning
- recent interest: Quantum Operations Research
- applications: Routing, Scheduling, WSN, Sonar networks, Warehouse management, ...

Teaching

Teaching programs

- Mathematical engineering
- Electronic design
- Production management
- Industrial engineering

Courses

- Intro OR/graph theory
- Mathematical programming
- Heuristics/metaheuristics
- Data analysis/management
- Large scale optimization

Level of students

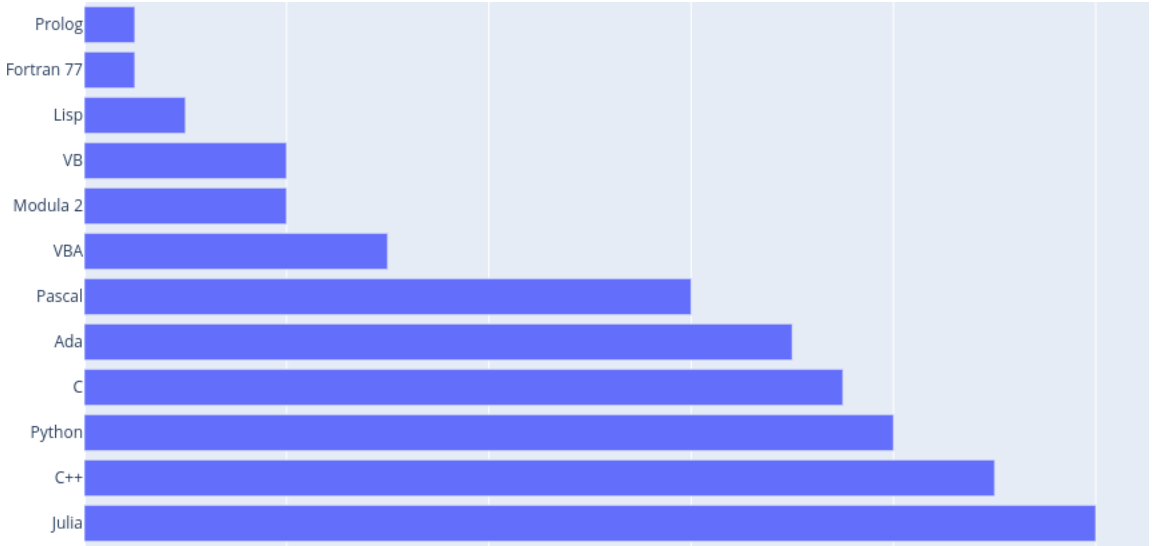
- Master students
- Engineer students
- PhD students

Type of students

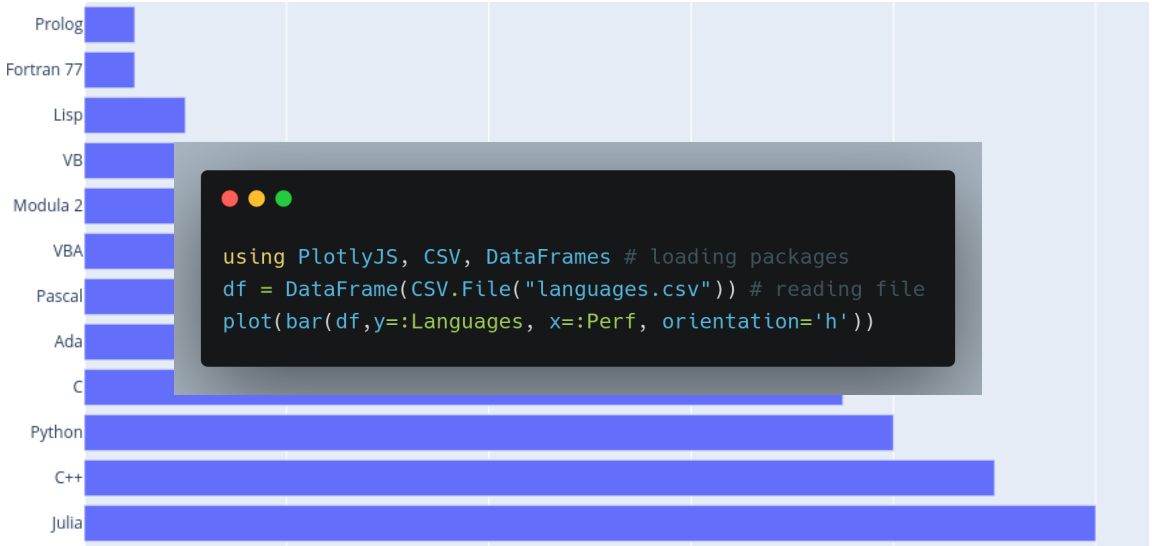
- Classical students
- Full distance learning
- Professional students

Not a specialist in Computer Science, but use programming languages in every class...

Programming history (proficiency)



Programming history (proficiency)



Why Julia?

4 reasons to learn julia today

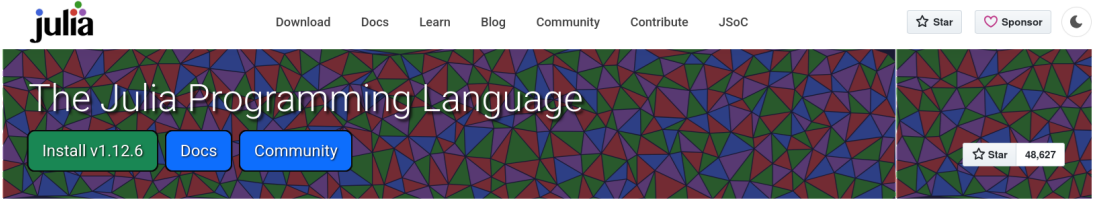
Speed as general as Python, as statistics-friendly as R, but as fast as C

Syntax dynamically-typed, closer to R and Python, easy to learn

Multiple dispatch Julia functions' ability to behave differently based on the types of arguments

OR friendly Julia's environment is well adapted to OR with rapid prototyping, speed, packages, link to other languages, ...

http://julialang.org



Fast

Julia was designed for high performance. Julia programs automatically compile to efficient native code via LLVM, and support multiple platforms.

Dynamic

Julia is dynamically typed, feels like a scripting language, and has good support for interactive use, but can also optionally be separately compiled.

Reproducible

Reproducible environments make it possible to recreate the same Julia environment every time, across platforms, with pre-built binaries.

Composable

Julia uses multiple dispatch as a paradigm, making it easy to express many object-oriented and functional programming patterns. The talk on the Unreasonable Effectiveness of Multiple Dispatch explains why it works so well.

General

Julia provides asynchronous I/O, metaprogramming, debugging, logging, profiling, a package manager, and the ability to build binaries.

Open source

Julia is an open source project with over 1,000 contributors. It is made available under the MIT license. The source code is available on GitHub. Julia has a welcoming community accessible to all backgrounds.

Julia REPL (Read Evaluate Print Loop)

Julia is Fast

Quick version

You call a function, Julia will specialize that call all the way down to its concrete types (thanks to multiple dispatch).

Let's see how Julia compare to other languages...

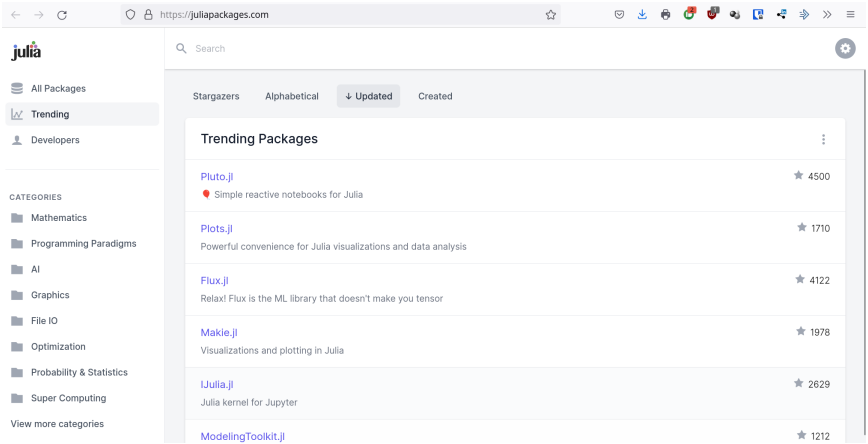
Julia is easy to understand

Intuitive and easy

- Julia is fast
- Julia is easier and cleaner than C++
- You can avoid the two-language problem

See `dataframes`, `arrays/vectors`, `functions` and `broadcast`...

Julia packages



Let's explore the packages `Graphs`, `PlotlyJS` and `JuMP`...

Conclusion

Cons.

- Julia is a young language
- Compile time latency may happen
- Ecosystem is immature
- Subtle syntax tricks

Pros.

- Julia is easy to learn
- Julia REPL is amazing
- Julia is fast
- Multithreading is easy
- I will you during the week :-)